

Program : Diploma in Tool and Die Engineering	
Course Code : 3111	Course Title: Moulds & Dies Technology
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To provide a thorough knowledge on plastic materials, their properties and processing methods in moulding industry
- To introduce different plastic processing techniques used in industry
- To give a basic information on the product design concept

Course Prerequisites:

Topic	Course code	Course name	Semester
Plastics		Applied Chemistry	1
Production processes		Production Engineering I & II	2 & 3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe plastic materials, properties and processing methods in moulding industry	15	Understanding
CO2	Apply feed and cooling system in mould	15	Applying
CO3	Explain the methods of plastic moulding other than injection moulding	14	Understanding
CO4	Recognize process description of die casting and the concept of product design	14	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3				2		
CO3	2				1		
CO4	2				1		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe plastic materials, properties and processing methods in moulding industry		
M1.01	Classify plastic materials and their properties	3	Understanding
M1.02	Differentiate between thermo and thermosetting plastics	4	Understanding
M1.03	Explain various moulding processes used in Industry	4	Understanding
M1.04	Describe working principle of injection moulding machine	4	Understanding

Contents:

Plastics in the mould industry - Definition of plastics - properties of plastics - application. - types

Thermo setting plastics - Thermo plastics - Examples - properties and uses of Nylon (Polyamide), PVC, PMMA, Acrylic, Polycarbonate, FRP, Polyoxymethylene (POM), Polystyrene (PS), ABS, Polypropylene (PP), Polyethylene (PE)

Plastic processes: Injection moulding, Compression moulding, Transfer moulding; Extruding; Casting; Calendaring; Fabrication methods-Sheet forming, Blow moulding, Laminating plastics (sheets, rods & tubes)

Injection moulding- working-principle-advantages and disadvantages

CO2	Apply feed and cooling systems in mould		
M2.01	Terminology used in mould making	3	Understanding
M2.02	Apply Feed system of mould	4	Applying
M2.03	Explain Ejection techniques	4	Understanding

M2.04	Apply cooling system and classification of mould	4	Applying
	Series Test - I	1	
Contents: Definition of mould -Terminology - mould materials Feed system - Runner - types of runner - Runner diameter calculation - Gate - types of gates - Gate calculation Ejection Techniques - Parting surface - venting - shrinkage - values of shrinkage - Factors governing shrinkage Temperature control - cooling - mould cooling (simple numerical problem) Side core and side cavity - Determine core and cavity dimensions - Internal undercuts - Two plate and three plate moulds - runner less moulds - multi daylight moulds - mould maintenance.			
CO3	Explain plastic processing techniques other than injection moulding process		
M3.01	Explain process of compression and transfer moulding	2	Understanding
M3.02	Explain process of extrusion and blow moulding	4	Understanding
M3.03	Describe the process of rotational moulding	4	Understanding
M3.04	Describe the process of calendaring, vacuum forming	4	Understanding
Contents: Compression moulding - application - temperature and pressure ranges - basic types of compression moulds - flash type mould - positive mould - landed positive mould - semi positive mould. Transfer moulding - application - advantages and limitations - types of transfer mould - pot type and plunger type Extrusion moulding - description of the process - application. Blow moulding - Description of the process - application. Rotational moulding - description of the process - application. calendaring - vacuum forming -filament winding - multicolor moulding			
CO4	Recognize process description of die casting and introduce the concept of product design		
M4.01	Explain die casting process	4	Understanding
M4.02	Classify Die casting machines - hot chamber and cold chamber	4	Understanding

M4.03	List Die casting alloys	2	Understanding
M4.04	Explain the basic concept of Product design	4	Understanding
	Series Test - II	1	

Contents:

Die casting - advantages of die casting techniques - principle advantages and limitations of pressure die casting process - die casting terminology - die casting alloys - zinc, aluminum, magnesium, copper etc. - composition - properties of different die casting alloys - die casting machines - hot chamber and cold chamber - comparison of hot chamber and cold chamber process - die casting dies

Product design - introduction - steps involved in product design - characteristics of product development - challenges of product development - phases involved in product development - steps involved in product planning process

Text/ Reference:

T/R	Book Title/Author
T1	Introduction to injection moulding - Pye
T2	The Mould Design Guide - Peter Jones
R1	Fundamentals of plastics mould design - Sanjay K Nayak - Tata Mac Graw
R2	Product design and development-Karl T. Ulrich ,Steven D. Eppinger-Mac Graw Hill
R3	Plastic technology handbook - Donald E Hudgin

Online Resources:

Sl.No	Website Link
1	https://myplasticmold.com/
2	https://www.starrapid.com/blog/the-ten-most-popular-plastic-injection-
3	https://plasticmold.net/